

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Alpha-Amino Acid Biosynthetic Process (GO:0006901)	0.26902178	10	3.222e-03	9.205e-01	CAD:346 BCAT2:565 SDS:598 SDSL:604 MTHFD1:847 ILVBL1:1549
Bile Acid And Bile Salt Transport (GO:0006901)	0.20679031	16	4.191e-03	9.205e-01	ABCC11:4 SLC10A6:86 SLC01A2:455 SLC27A5:1052 SLC01B1:1485 SLC10A2:1495
Cellular Response To Acetylcholine (GO:0006901)	-0.30102484	10	9.804e-04	9.205e-01	CHRN2:60 CHRM3:241 CHRNA3:347 LVH3:375 CHRN2:603 LYPD1:2204
Chromatin Remodeling (GO:0006338)	0.06047363	200	3.297e-03	9.205e-01	HIRA:7 CHD8:27 GATAD2A:43 DAXX:80 RNF168:116 DDX12:145
Cilium-Dependent Cell Motility (GO:00602)	-0.20843465	17	2.931e-03	9.205e-01	DNAH2:9 TEK1:157 DNAH17:186 TEK2:1032 GAS8:1325 TEK14:1573
Glycerophospholipid Metabolic Process (GO:0006901)	0.22247174	56	1.501e-03	9.205e-01	PLA2G2D:205 ABHD3:368 PLA2G7:432 PIGG:586 SMPD4:707 PIGV:769
Mitochondrial Cytochrome C Oxidase Assem (GO:0006901)	-0.21434926	15	4.038e-03	9.205e-01	FASTKD3:648 TIMM21:735 COA1:1637 COX16:1767 SURF1:1958 COA8:2061
Muscle Contraction (GO:0006936)	-0.09908076	87	1.422e-03	9.205e-01	MYH13:18 TACR2:22 CHRN2:60 ARHGEF11:147 MYH8:352 CHRN2:1383
Negative Regulation Of CD8-positive, Alp (GO:0006901)	-0.34738851	6	3.210e-03	9.205e-01	ZBTB78:227 SOCS1:511 HFE:1822 VSIR:1946 SLC4A2:2454 DAPL1:7900
Neuropeptide Signaling Pathway (GO:00072)	-0.11308371	63	1.927e-03	9.205e-01	NPBW2:75 SSTR4:256 CYSLTR2:353 OPRK1:396 RXFP3:397 GALT1:440
Nuclear Pore Organization (GO:0006999)	0.21692473	14	4.955e-03	9.205e-01	NUP98:502 NDC1:514 NUP133:946 NUP107:1042 NUP205:1069 NUP54:1249
Positive Regulation Of Cell Cycle Process (GO:0006901)	0.08631847	105	2.316e-03	9.205e-01	KIF14:54 ZNF16:125 EGF:181 KIF20B:276 STL:291 SMC2:311
Positive Regulation Of Chromosome Segreg (GO:0006901)	0.21401943	17	2.254e-03	9.205e-01	SMC2:311 NUMA1:606 CDC8:885 NCAPH:1298 RCC2:1413 SMC4:1577
Positive Regulation Of Histone Deacetyla (GO:0006901)	0.26303025	12	1.607e-03	9.205e-01	SMARCA5:399 PRKD2:1177 VEGFA:1244 SIRT1:1319 BAZ2A:1376 SIRT6:1400
Positive Regulation Of Protein Deacetyla (GO:0006901)	0.24504001	13	2.222e-03	9.205e-01	SMARCA5:399 FNTA:887 IFNG:1099 SIRT1:1319 BAZ2A:1376 NIPBL:2199
Positive Regulation Of Transcription By (GO:0006901)	0.29906611	12	3.346e-04	9.205e-01	CHD8:27 DDX21:145 ERCC26:204 SMARCA5:399 ELL:471 ZC3H8:725
Positive Regulation Of Tyrosine Phosphor (GO:0006901)	0.11929692	49	3.890e-03	9.205e-01	TNF:259 IL23R:345 IGF1:669 PARG:686 CNTF:754 CSF2:789
Preassembly Of GPI Anchor In ER Membrane (GO:0006901)	0.3098785	7	4.535e-03	9.205e-01	PIGG:586 PIGV:769 PIGB:788 PIGN:932 PIGW:2007 PIGL:8068
Regulation Of Rho Protein Signal Transdu (GO:0006901)	-0.10368282	68	3.141e-03	9.205e-01	EPS8L3:57 RIPOR2:104 KANK1:117 ARHGEF2:221 GPR35:441 KANK2:601
Regulation Of Cell Cycle (GO:0051726)	-0.05211947	260	3.970e-03	9.205e-01	BIRC3:45 RNF167:132 BIRC7:136 TBGR4:169 FIGNL1:292 ZBTB17:380
Regulation Of Chromosome Separation (GO:0006901)	0.27222758	9	4.685e-03	9.205e-01	SMC2:311 NUMA1:606 NCAPH:1298 SMC4:1577 NCAPG2:1675 NCAPD2:2960
Regulation Of Interleukin-13 Production (GO:0006901)	0.02472262	9	4.383e-03	9.205e-01	IRF4:529 SPHK2:530 IL4:952 TNFRSF21:1288 TSLP:2022 LEF1:2815
Regulation Of Neuron Projection Developm (GO:0006901)	-0.06774229	165	2.752e-03	9.205e-01	MINAR1:7 CAMK2B:66 KANK1:117 ITGA6:199 ROR2:308 TENM3:434
Regulation Of Transcription By RNA Polym (GO:0006901)	0.20191780	22	1.046e-03	9.205e-01	CHD8:27 DDX21:145 ERCC6:204 DHX36:258 SMARCA5:399 ELL:471
Serine Family Amino Acid Biosynthetic Pro (GO:0006901)	0.21722738	9	4.839e-03	9.205e-01	CTH:608 SERINC3:745 MTHFD1:847 SRR:1930 PSHP:2220 SERINC5:2248
sRNA Transcription By RNA Polymerase II (GO:0006901)	0.31382881	7	4.036e-03	9.205e-01	SNAPC1:277 ELL:471 ZC3H8:725 ICE2:762 SNAPC5:2371 ELL:471
snRNA Transcription By RNA Polymerase I (GO:0006901)	0.43514501	4	2.577e-03	9.205e-01	SNAPC1:277 ZC3H8:725 ICE2:762 SNAPC5:2371 NA NA
Steroid Hormone Biosynthetic Process (GO:0006901)	0.21240953	16	3.269e-03	9.205e-01	HSD17B6:146 CYP17A1:593 HSD17B3:600 HSD17B2:613 HSD17B11:893 CYP11A1:1087
Steroid Metabolic Process (GO:0008202)	0.11549574	61	1.254e-03	9.205e-01	HSD17B6:146 CYP51A1:314 HSD11B1:413 CYP17A1:593 HSD17B3:600 HSD17B2:613
Glut Movement Involved In Cell Motilit (GO:0006901)	-0.30353125	7	5.419e-03	9.450e-01	TEKT1:157 TEKT2:1023 GAS8:1325 TEKT4:208 TEKT5:2367 RSPH9:7900
Mitotic G1 DNA Damage Checkpoint Signali (GO:0006901)	-0.15646549	26	5.767e-03	9.450e-01	SDE2:264 MUC1:377 GML:389 PML:593 PLK3:753 GTS1:952
Negative Regulation Of Endothelial Cell (GO:0006901)	0.15949878	25	5.788e-03	9.450e-01	NFE2L2:160 KRIT1:942 IL4:952 ICAM1:1194 FGG:1294 TEK:1440
Neutral Amino Acid Transport (GO:0015804)	0.13712107	34	5.678e-03	9.450e-01	SLC43A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC36A1:338 SLC6A9:362
DNA Integrity Checkpoint Signaling (GO:0006901)	0.11558151	36	1.645e-02	9.862e-01	TP53BP1:94 ERCC26:204 CDC45:603 PARG:686 CLSPN:864 CDC6:885
DNA Replication Initiation (GO:0006270)	0.16029282	20	1.310e-02	9.862e-01	ORC1:65 ORC3:597 CDC45:603 ORC2:1104 LRWD1:1396 PRIM2:1588
L-phenylalanine Catabolic Process (GO:0006901)	0.31263606	5	1.031e-02	9.862e-01	TAT:592 HGD:715 QDDR:1714 IL4I1:2352 GSTZ1:8068 NA
L-phenylalanine Metabolic Process (GO:0006901)	0.31263606	5	1.031e-02	9.862e-01	TAT:592 HGD:715 QDDR:1714 IL4I1:2352 GSTZ1:8068 NA
L-serine Metabolic Process (GO:0006563)	0.24584037	10	7.106e-03	9.862e-01	SDS:598 SDSL:604 SERINC3:745 CBS1:692 SRR:1930 PSHP:2220
Anterograde Trans-Synaptic Signaling (GO:0006901)	-0.05418828	180	1.242e-02	9.862e-01	MINK1:56 CHRN2:60 HTR3B:177 CHRM3:241 CHRNA3:347 CHRN2:1383
Aromatic Compound Catabolic Process (GO:0006901)	0.26068214	7	1.692e-02	9.862e-01	CYP2F1:30 TET1:719 PON3:900 PDXP:1354 PON1:8068 PON2:8068

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO (GO:0006901)	-0.17013844	127	3.725e-11	2.417e-07	LAMA5:61 HELZ2:64 NPBW2:75 BIRC7:136 GAT5:200 ABHD16B:224
HAMAL_APOPTOSIS_VIA_TRAIL_UP (GO:0006901)	0.07614772	591	3.180e-10	1.032e-06	TASOR:14 SPG11:25 SMCHD1:35 CEP192:48 KIF14:54 NEK1:64
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN (GO:0006901)	0.06143360	809	3.778e-09	8.171e-06	TMEM131L:6 HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SPG11:25
NIKOLSKY_BREAST_CANCER_Q21_AMPLICON (GO:0006901)	-0.28843801	33	9.833e-09	1.595e-05	DCST1:51 ADAM15:99 MTX1:134 PBXIP1:142 RUSC1:181 ZBTB78:227
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN (GO:0006901)	0.07693100	441	3.458e-08	4.487e-05	HIRA:7 MAN2A1:13 TASOR:14 UBR2:18 SMCHD1:35 ZNF292:40
-0.21378117 (GO:0006901)	0.21378117	46	5.299e-07	5.730e-04	PIEZO1:11 CPNE7:192 ZC3H18:229 ANKRD11:334 PABPN1L:456 C8FA2T3:522
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON (GO:0006901)	-0.37576264	14	1.127e-06	1.044e-03	COL6A1:73 COL6A2:87 SPATC1L:165 PCNT:236 DIP2A:606 COL18A1:808
MARTENS_TRETINOIN_RESPONSE_UP (GO:0006901)	-0.05423703	675	1.803e-06	1.462e-03	B3GN76:19 SMTNL2:29 MMEL1:30 CARD14:41 C1QTNF8:43 COL6A1:73
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON (GO:0006901)	-0.34442946	15	3.863e-06	2.703e-03	SFSWAP:17 GALTNR:616 NOC4L:631 PUS1:636 GOLGA3:676 EPO40:682
REACTOME_SIGNALING_BY_GPCR (GO:0006901)	-0.05401748	623	4.737e-06	2.703e-03	TACR2:22 GPR132:47 CAMK2B:66 NPBW2:75 BDKRB1:86 ARHGEF11:147
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT (GO:0006901)	-0.07733324	296	4.999e-06	2.703e-03	TACR2:22 GPR132:47 NPBW2:75 BDKRB1:86 GPR68:150 CHRM3:241
DAZARD_RESPONSE_TO_UV_NHEK_DN (GO:0006901)	0.07996617	280	4.858e-06	2.703e-03	TMEM131L:6 WDFY3:37 KIF14:54 AKAP9:82 DOP1A:95 DST:106
WP_GPCRS_CLASS_A_RHODOPSLIKE (GO:0006901)	-0.08442194	211	2.453e-05	1.224e-02	NPBW2:75 BDKRB1:86 GPR68:150 OR6B1:207 CHRM3:241 SSTR4:256
STARK_HYPOCAMPUS_22Q11_DELETION_DN (GO:0006901)	0.25685484	22	3.042e-05	1.410e-02	HIRA:7 PRODH:234 TANGO2:283 DGCRC6:343 DGCRC2:538 COMT:948
HSIAO_LIVER_SPECIFIC_GENES (GO:0006901)	0.08490669	199	3.749e-05	1.621e-02	GSTO1:20 F11:57 SERPIND1:67 ADH1A1:85 ALB:142 MGST1:216
PEREZ_TP63_TARGETS (GO:0006901)	-0.06475642	313	8.548e-05	3.262e-02	CEMP1:6 NLRP1:49 KLK15:67 SEMA4D:155 APC2:212 EVPL:226
SCHLOSSER_SERUM_RESPONSE_DN (GO:0006901)	0.04662107	618	8.344e-05	3.262e-02	TASOR:14 UBR2:18 HEXB:21 ICAM2:28 ASMTL:29 PTPRC:47
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS (GO:0006901)	0.10797444	101	1.792e-04	6.459e-02	SLC43A1:32 SLC6A6:143 SLC7A6:157 SLC36A2:219 SLC26A3:273 SLC36A1:338
PYEON_CANCER_HEAD_AND_NECK_VS_CERVICAL_U (GO:0006901)	0.08397830	166	1.930e-04	6.591e-02	NR1D2:91 WDR76:134 SLF1:196 NUP210:202 HLF:206 TUG1:243
WP_2Q112_COPY_NUMBER_VARIATION_SYNDROME (GO:0006901)	0.18366797	34	2.108e-04	6.837e-02	ARID5A:98 CNMNA:178 MMS1:290 CNMNA:178 KANSL3:474 COL2A1:503
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE (GO:0006901)	0.04466657	586	2.384e-04	7.366e-02	KIF14:54 WDR76:134 APC:147 AGGF1:167 SLF1:196 HLF:206
REACTOME_GPCR_LIGAND_BINDING (GO:0006901)	-0.05305992	398	2.553e-04	7.370e-02	TACR2:22 GPR132:47 NPBW2:75 BDKRB1:86 GPR68:150 CHRM3:241
DUTERTE_BESTADIOL_RESPONSE_24HR_UP (GO:0006901)	0.06065809	308	2.613e-04	7.370e-02	DEPTOR:88 WDR76:134 TET2:179 NLRP5:425 NUP210:202 RAP1GAP:232
ZHANG_BREAST_CANCER_PROGENITORS_UP (GO:0006901)	0.05210130	408	3.214e-04	8.687e-02	PROM1:1 SMCHD1:35 CEP192:48 ANAPC1:62 NEK1:64 AKAP9:82
WP_PEPTIDE_GPCRS (GO:0006901)	-0.12205469	70	4.165e-04	1.055e-01	TACR2:22 BDKRB1:86 SSTR4:256 OPRK1:396 GALT1:440 FSHR:490
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME (GO:0006901)	0.10012894	104	4.230e-04	1.055e-01	HIRA:7 CLTC1:19 PAKA:36 SERPIND1:67 SLC7A4:120 DEPDC5:136
DODD_NASOPHARYNGEAL_CARCINOMA_DN (GO:0006901)	0.03016670	1204	4.957e-04	1.191e-01	GSTO1:20 CEP192:48 KIF14:54 ANAPC1:62 EMLIN:266 GSER1:127
FISCHER_DREAM_TARGETS (GO:0006901)	0.03346534	865	9.215e-04	2.135e-01	MTBP:5 SMCHD1:35 GATAD2A:43 FNBP4:44 CEP192:48 KIF14:54
REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR (GO:0006901)	-0.07399589	168	9.673e-04	2.164e-01	TACR2:22 NPBW2:75 BDKRB1:86 SSTR4:256 C3:296 NPW:340
ASGHARZADEH_NEUROBLASTOMA_POOR_SURVIVAL (GO:0006901)	-0.15004354	39	1.188e-03	2.569e-01	ADCY1:495 C10orf35:576 GATAD2B:775 HAP1:904 H2AZ2:999 TMOD2:1054
CAIRO_HEPATOBLASTOMA_DN (GO:0006901)	0.06121765	234	1.285e-03	2.688e-01	TMEM131L:6 F11:57 OGDHL:93 CA2:140 HSD17B6:146 LEPR:190
BENPORATH_ES_WITH_H3K27ME3 (GO:0006901)	-0.03042522	976	1.434e-03	2.745e-01	CEMP1:6 PLEC:12 BNC1:28 ANKRD35:31 CAMK2B:66 LONRF3:85
DAZARD_UV_RESPONSE_CLUSTER_G6 (GO:0006901)	0.07923190	137	1.382e-03	2.745e-01	KIF14:54 DOP1A:95 TUT2:156 UBR5:166 ZNF451:187 OTUD4:201
TOYOTA_TARGETS_OF_MIR34B_AND_MIR34C (GO:0006901)	0.04654378	401	1.439e-03	2.745e-01	PPP4R1:10 KIF14:54 F4R1:128 DDX21:145 NUP210:202 RAP1GAP:232
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND (GO:0006901)	0.36837516	6	1.778e-03	3.105e-01	CA2:140 C4A:195 SLC4A1:1012 RHAG:1018 CA1:2187 AQP1:8068
DEURIG_T_CELL_PROLIFEROUSCYTIC_LEUKEMIA_DN (GO:0006901)	0.05567390	266	1.818e-03	3.105e-01	MAN2A1:13 TASOR:14 ABCB1:15 ZNF292:40 PTPRC:47 PAJ2:71
KINSEY_TARGETS_OF_EWSR1_FUSION_UP (GO:0006901)	0.02751912	1172	1.699e-03	3.105e-01	PROM1:1 KIF14:54 ORC1:65 RADX:81 WDR76:134 DDX21:145
KEGG_STEROID_HORMONE_BIOSYNTHESIS (GO:0006901)	0.17034477	28	1.811e-03	3.105e-01	HSD17B6:146 HSD11B1:413 AKR1D1:460 CYP17A1:593 HSD17B3:600 HSD17B2:613
MEISSNER_NPC_HCP_WITH_H3K27ME3 (GO:0006901)	-0.10326135	76	1.866e-03	3.105e-01	GAT5:200 BAGALN2:455 SFRP5:479 TCF15:770 GRM7:827 HCK:906
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON (GO:0006901)	-0.08996657	99	1.988e-03	3.225e-01	C1QTNF8:43 EME2:286 NPW:340 UNKL:605 SYNGR3:738 NME3:781

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hypocalcemia (GO:0006901)	0.20366287	45	2.311e-06	2.267e-02	HIRA:7 CASR:63 IFT122:154 UBR1:214 DGCRC6:343 CLDN16:423
DiGeorge Syndrome (GO:0006901)	0.15568988	61	2.646e-05	9.254e-02	HIRA:7 SPECCL1:138 ALB:142 PRODH:234 TJP1:282 DGCRC6:343
Nephrocalcinosis (GO:0006901)	0.16482634	54	2.830e-05	9.254e-02	CLCNKB:41 NSD1:42 CASR:63 SLC26A3:273 CLDN16:423 SLC12A1:431
Tetany (GO:0006901)	0.25943809	21	3.874e-05	9.500e-02	HIRA:7 CASR:63 CLDN16:423 SLC12A1:431 TRPM6:624 GCM2:629
Shprintzen syndrome (GO:0006901)	0.20149727	33	6.218e-05	1.220e-01	HIRA:7 PRODH:234 DGCRC6:343 DGCRC6L:366 CHRNA7:511 DGCRC2:538
Specific learning disability (GO:0006901)	0.12207327	86	9.300e-05	1.521e-01	HIRA:7 SPG11:25 NSD1:42 SLL1:165 PAKM:279 DGCRC6:343
22q11 partial monosomy syndrome (GO:0006901)	0.33633222	11	1.123e-04	1.574e-01	HIRA:7 PAKA:36 JMJD1C:780 COMT:948 GNB1L:1306 ARVCF:1586
Pain, Postoperative (GO:0006901)	0.22719548	21	3.140e-04	3.850e-01	ABCB1:15 TNF:259 TRPV1:263 SCN1LA:367 AP3B1:644 PTPRC:47
Anemia, hereditary spherocytic hemolytic (GO:0006901)	0.42171455	5	1.091e-03	6.798e-01	ANK1:89 EPB42:568 SLC4A1:1012 SPTB:1657 SPTAL1:2807 NA
Acanthosis (GO:0006901)	0.30612581	10	8.023e-04	6.798e-01	CAD34:66 KEL:409 VPS13A:420 XK:824 PIK3C2A:916 MTPP:1120
Cervical Squamous Intraepithelial Neopla (GO:0006901)	0.14217909	43	1.264e-03	6.798e-01	APC:147 LFN:259 MAM:135 FBXW7:525 MET:558 LAMA3:617
Idiopathic hypercalciuria (GO:0006901)	0.35501167	7	1.143e-03	6.798e-01	CASR:63 SLC12A1:431 VDR:964 SLC12A3:1299 CLCN5:2330 TRPV5:2880
Increased calcium level in kidney (GO:0006901)	0.16711728	33	8.965e-04	6.798e-01	NSD1:42 CASR:63 CLDN16:423 SLC12A1:431 COL4A3:456 GCM2:629
Occipital myelomeningocele (GO:0006901)	0.30946896	9	1.305e-03	6.798e-01	HIRA:7 JMJD1C:780 COMT:948 ARVCF:1586 TBX1:2367 RNF81:2560
Polyglandular Type I Autoimmune Syndrome (GO:0006901)	0.18770421	30	7.078e-04	6.798e-01	VCAM1:24 CASR:63 TNF:259